

CS & IT ENGINEERING

DATA STRUCTURE

Hashing



JOIN TELEGRAM TO ENROLL

SHAIENDRA SIR

Consider a hash table of size seven, with starting index zero, and a hash function $(3x + 4) \bmod 7$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '-' denotes an empty location in the table.

- A 8, -, -, -, -, -, 10
- ☒ B 1, 8, 10, -, -, -, 3
- C 1, -, -, -, -, -, 3
- D 1, 10, 8, -, -, -, 3

$$h(x) = (3x + 4) \% 7$$

$$x = 1$$

$$h(1) = 7 \% 7 = 0$$

$$h(3) = (3 \times 3 + 4) \% 7 = 13 \% 7 = 6$$

$$h(8) = (3 \times 8 + 4) \% 7 = 28 \% 7 = 0$$

$$\left[\begin{array}{c} 1 \\ 0 \end{array} \quad \begin{array}{c} 8 \\ 1 \end{array} \quad \begin{array}{c} - \\ 2 \end{array} \quad \begin{array}{c} - \\ 3 \end{array} \quad \begin{array}{c} - \\ 4 \end{array} \quad \begin{array}{c} - \\ 5 \end{array} \quad \begin{array}{c} 3 \\ 6 \end{array} \right]$$

An advantage of chained hash table (external hashing) over the open addressing scheme is

- A. Worst case complexity of search operations is less
- B. Space used is less
- ☒ C. Deletion is easier
- D. None of the above

Quadratic Probing: $HF(key) = key \bmod m$, $m = 10$ (0-9)

$$QP(Key, i) = (HF(Key) + C_1 * i + C_2 * i^2) \bmod m$$

Keys: 25, 98, 57, 75, 97, 18, 78, 46

$C_1=1, C_2=1, i=0, 1, 2, 3, \dots, 9$ (m-1)

$$QP(25, 0) = (HF(25) + 1 \times 0 + 1 \times 0^2) \bmod 10$$

$$= (25 \bmod 10) \bmod 10 = 5$$

$$QP(98, 0) = 8 \quad QP(97, 0) = 7 \text{ collision}$$

$$QP(57, 0) = 7 \quad QP(97, 1) = 9$$

$$QP(75, 0) = 5 \text{ collision} \quad QP(18, 0) = 8 \text{ collision}$$

$$QP(75, 1) = (5 + 1 \times 1 + 1 \times 1^2) \bmod 10 \quad QP(18, 1) = 0$$

$$= 7 \text{ collision}$$

$$QP(75, 2) = (5 + 2 \times 1 + 2^2) \bmod 10 = 1$$

0	18
1	75
2	NULL
3	NULL
4	78
5	25
6	46
7	57
8	98
9	97

Total Collision = 6
Ans

HashTable

Ex:- $key(n) = 41, 60, 82, 93, 70, 80, m=10, HF(key) = key \bmod m$

$$QP(key, i) = (HF(key) + C_1 i + C_2 i^2) \bmod m; i=0, 1, 2, \dots, m-1$$

$$QP(70, 0) = 0 \text{ Collision}$$

$$QP(70, 1) = 2 \text{ Collision}$$

$$QP(70, 2) = 0 + 2 + 4 = 6 \checkmark$$

$$QP(80, 0) = 0 \text{ Collision}$$

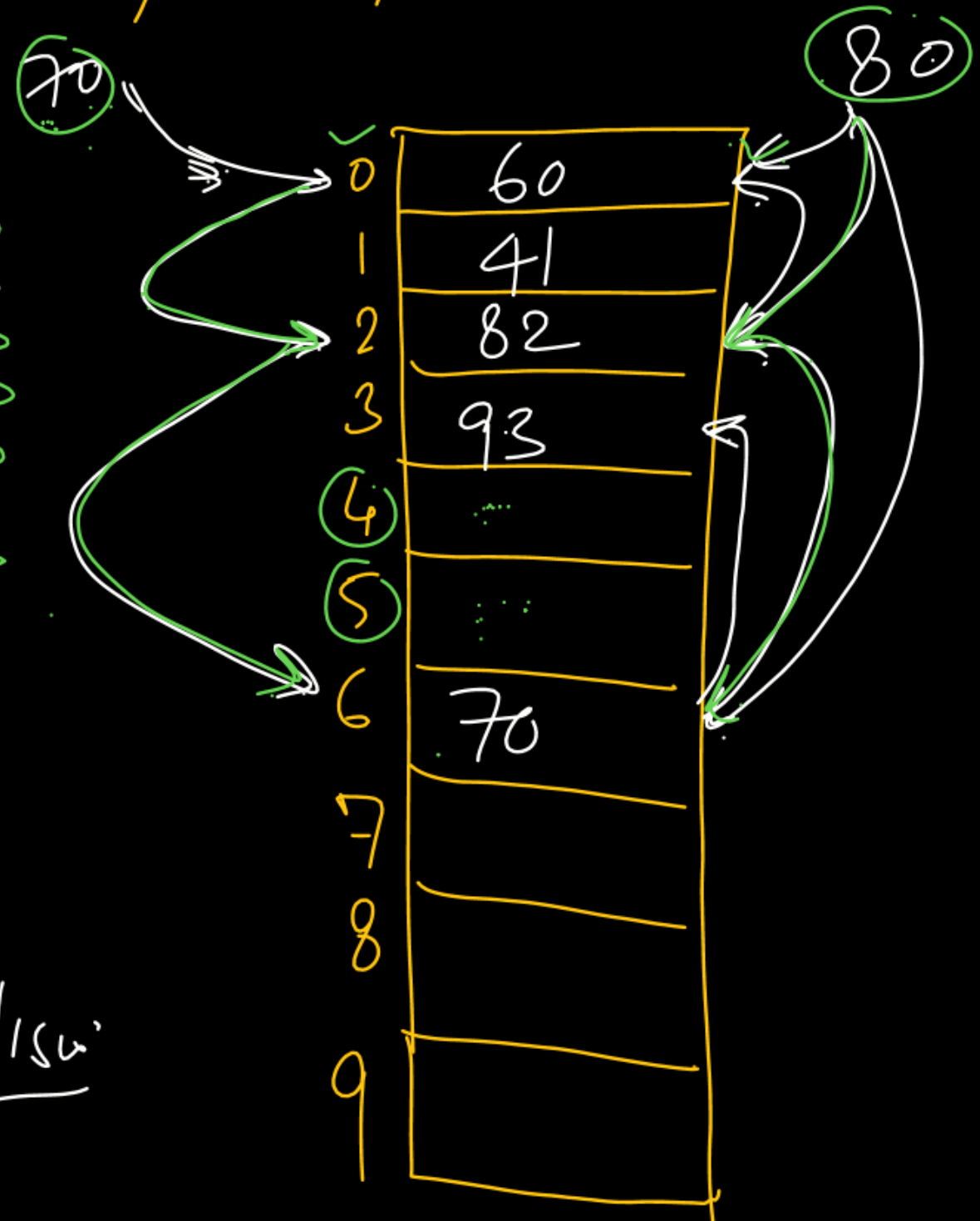
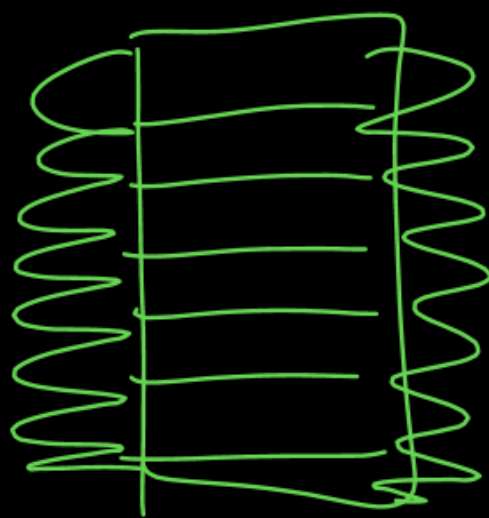
$$QP(80, 1) = 2 \text{ Collision}$$

$$QP(80, 2) = 6 \text{ Collision}$$

$$QP(80, 3) = 0 + 3 + 9 = 12 \bmod 10 = 2 \text{ Collision}$$

$$QP(80, 4) = 0 + 4 + 16 = 20 \bmod 10 = 0 \text{ Collision}$$

$$QP(80, 5) = 0 + 5 + 25 = 30 \bmod 10 = 0 \text{ Collision}$$



Secondary clustering:-

- 1) if two keys are mapped onto same starting location in the hash Table then they both follow the same path in the Quadratic manner because of its search time complexity will increase
- 2) Secondary clustering smaller than Primary clustering, because of this reason average searching time decreases
- 3) To avoid this problem Double Hashing will be used
- 4) Deletion will create the problem to other keys (Just like Linear Probing) But handle by special symbol (#) just like Linear Probing

TC	BC	WC
Insertion	$O(1)$	$O(m)$
Searching	$O(1)$	$O(m)$
Deletion	$O(1)$	$O(m)$

Double Hashing:-

$$HF_1(\text{key}) = \text{key} \bmod m, \quad m = 10 \text{ (0-9)}$$

$$HF_2(\text{key}) = 1 + (\text{key} \bmod (m-2))$$

$$DH(\text{key}, i) = (HF_1(\text{key}) + i \cdot HF_2(\text{key})) \bmod m$$

$i = 0, 1, 2, 3, \dots, 9(m-1)$

Ex:- Keys(n) = ~~25~~, ~~98~~, ~~57~~, ~~75~~, ~~97~~, ~~18~~, ~~78~~, ~~65~~, ~~46~~, Total Collision?

$$DH(25, 0) = 5$$

$$DH(98, 0) = 8$$

$$DH(57, 0) = 7$$

$$DH(75, 0) = 5 \text{ collision}$$

$$\begin{aligned} DH(75, 1) &= (5 + 1 \times (1 + 75 \bmod 8)) \bmod 10 \\ &= (5 + 4) \bmod 10 = 9 \end{aligned}$$

$$DH(97, 0) = 7 \text{ collision}$$

$$DH(97, 1) =$$

0	
1	97
2	78
3	65
4	18
5	25
6	46
7	57
8	98
9	75

Hash Table

Q.65 In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i -th probe in the open address table is $[h_1(k) + i h_2(k)] \bmod m$. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is 10 Ans. (Answer in integer)

GATE-2025

$$H(k, i) = [h_1(k) + i h_2(k)] \bmod 11, i = 0, 1, 2, \dots, 10$$

$$H(63, 0) = 63 \bmod 11 = 8$$

$$H(50, 0) = 50 \bmod 11 = 6$$

$$H(25, 0) = 3$$

$$H(79, 0) = 2$$

$$H(67, 0) = 1$$

$$H(24, 0) = 2 \text{ Collision}$$

$$H(24, 1) = [2 + 1 \times (1 + 24 \bmod 7)] \bmod 11$$

$$= 6 \text{ Collision}$$

$$H(24, 2) = [2 + 2 \times 4] \bmod 11$$

$$= 10 \bmod 11 = 10$$

0	
1	67
2	79
3	25
4	
5	
6	50
7	
8	63
9	
10	24

Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 23$, and the secondary hash function is $h_2(k) = 1 + (k \bmod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k=90$ is _____.

$$m = 23$$

$$h_1(k) = k \bmod 23$$

$$h_2(k) = 1 + (k \bmod 19)$$

$$i = 1$$

$$DH(90, 1) = (h_1(k) + i * h_2(k)) \bmod 23$$

$$= [90 \bmod 23 + 1 * (1 + 90 \bmod 19)] \bmod 23$$

$$= [21 + 1 * (1 + 14)] \bmod 23$$

$$= 36 \bmod 23 \Rightarrow \textcircled{13} \text{ Ans}$$

$$\Rightarrow \textcircled{13}$$

0	
1	
2	
3	
⋮	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	

NOTE:-

1) The expected no. of Probes in an Unsuccessful Search of open Addressing Technique is $= \frac{1}{1-\alpha}$, where α is load factor

2) The expected no. of Probes in Successful search of open Addressing Technique is $= \frac{1}{\alpha} \log \frac{1}{1-\alpha}$, where α is load factor ($\alpha = \frac{n}{m}$)



Shailendra Singh GATE CSE/DA

THANKYOU



GeeksforGeeks GATE